

Proactive Support through floating window in Information Foraging with User Attention Modeling via Eye-tracking

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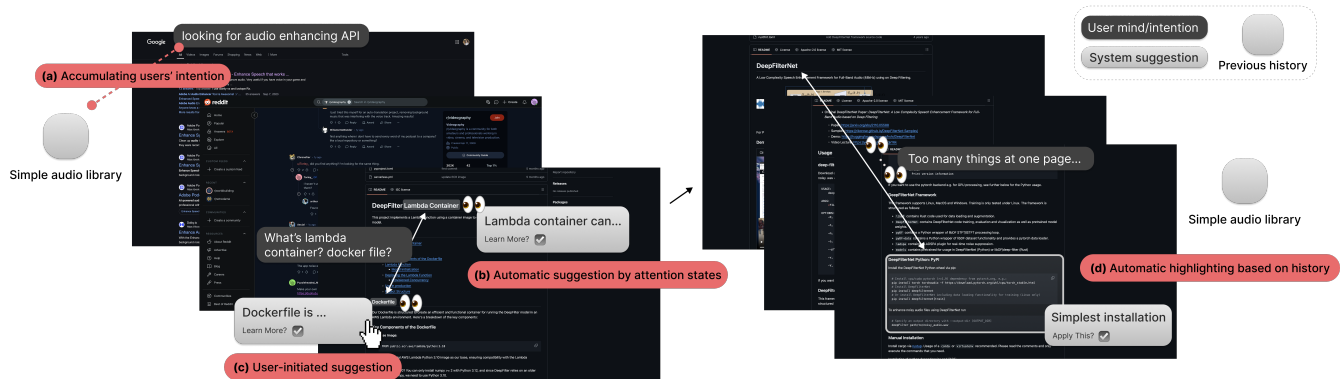


Figure 1: A proactive support system leveraging user attention modeling via eye-tracking in information foraging tasks. (a) The system accumulates user intention across different search and browsing contexts. (b) Based on attention states, the system proactively suggests relevant content. (c) Users can trigger suggestions through interactions. (d) The system automatically highlights previously attended content to reduce cognitive overload. The floating support windows adaptively respond to inferred user goals, aiding efficient navigation in complex information spaces.

Abstract

In complex online information foraging tasks, users often face cognitive overload due to fragmented interfaces and the limits of working memory. While large language models (LLMs) offer content-based assistance, they often miss users' true information needs by relying solely on browsing history. We introduce an attention-driven system that models user gaze as a grounding signal to infer cognitive states and evolving mental context. Leveraging real-time eye-tracking, our system delivers proactive interventions—contextual zoom-in, information pruning, and distraction recovery—through a lightweight floating UI. Designed to be non-intrusive and seamlessly integrated into users' browsing workflow, the interface enhances understanding without disrupting task flow or requiring explicit prompts. By accumulating attention over time, the system builds a dynamic knowledge graph to support multi-session exploration. This position paper shows how combining attention modeling with

proactive support design can reduce cognitive load, support comprehension, and enable more intuitive, context-sensitive information foraging.

CCS Concepts

• **Human-centered computing;**

Keywords

Human-AI Collaboration, User Interface, Sensemaking, Generative AI, Large Language Models, Eye-tracking, Multi-Modality, Cognitive Modeling

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1 Introduction

In an online environment, users frequently engage in information tasks that involve exploring, understanding, and synthesizing unfamiliar domains or concepts to develop a comprehensive understanding or make informed decisions. This iterative process exposes users to a large volume of information, primarily accessed through web browsing, which requires repeatedly navigating between multiple web pages and windows. Figure 1 illustrates this scenario, showing how users visit various websites while searching for an

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audio resolution enhancing API with a specific purpose. A key challenge in this process is the limited capacity of working memory when processing unfamiliar information [36, 48], compounded by extraneous cognitive load [27] introduced by disjointed web interfaces that fragment information across stacked windows and interrupt users' non-linear foraging patterns.

To address these challenges, prior research has proposed systems that proactively re-organize or summarize content [12, 22, 32?, 33], often layered on top of the current web interface. Other studies support information foraging in multi-criteria decision-making [10, 31, 32] or externalize users' mental contexts [11, 29] using similar methods. Many of these systems increasingly incorporate large language models (LLMs) to support content understanding and retrieval.

However, simply tuning LLMs based on users' browsing history is often insufficient. Users frequently engage in "selective reading" [54, 55], focusing only on parts of a page they find relevant while ignoring the rest. This creates a disconnect: what users visit isn't always what they read, and what they read isn't always what they understand. As a result, LLMs that rely solely on browsing logs may misidentify users' true information needs, resulting in either irrelevant suggestions [46] or misleading over-accommodations [7, 43–45].

To bridge this gap, we propose modeling user attention as an act of grounding [13]. Understanding user intention and status can make LLMs serve more contextual and helpful support. For effective attention modeling, we utilize eye-tracking as a foundation. Gaze is a key nonverbal signal for intent [3, 17, 25], and prior work has shown its utility in understanding user behaviors [14] and cognitive states [21, 30, 51, 52]. By processing gaze data into attention states, our system can deliver more context-sensitive interventions.

Building on this modeling method, we introduce an integrated system that combines attention-driven proactive support with knowledge graph construction. As shown in Figure 1(a), our system accumulates attention history over time to support context-aware interventions across sessions. The floating window interface delivers timely suggestions—such as contextual explanations or content highlights—without disrupting workflow or requiring prompt crafting. Unlike previous methods, our approach filters and structures information using real-time attention data rather than static browsing history, enabling more relevant and cognitively aligned support.

We implement three proactive interventions based on user attention states: (1) *Contextual Zoom-in*, triggered by sustained fixations, provides compact explanations via floating cards; (2) *Information Pruning*, activated by dispersed gaze patterns, highlights content relevant to prior attention; and (3) *Distraction Recovery*, which helps users regain focus after detecting repetitive or aimless navigation.

In summary, this paper explores how attention modeling via eye tracking can enhance web interfaces for information foraging. Our system combines proactive support through floating UI elements with a dynamic knowledge graph grounded in attention data. By doing so, we address the limitations of working memory and extraneous cognitive load while improving long-term information retrieval and understanding in multi-session exploration.

2 Related Works

2.1 Sensemaking in Online Information Foraging

Existing research on supporting sensemaking in online information foraging has addressed these challenges from various perspectives. Some studies [4, 18, 38, 42] are focused on information collection step, providing an overview of the information space not to be confused by encountering unfamiliar things. For the organizing information step, externalizing their mental contexts, such as intention or interests, instantaneously was effective without any methods out of the web browsers to arrange their understanding, which occurs in additional steps [11, 29]. Research addressing the online decision-making process [10, 31, 32] handled the issue from multiple criteria and options existing. Other studies [12, 15, 22, 32, 33] proposed proactive approaches to structure the information by tracking the contents of interests. Most of these tasks have approached representation by providing re-organized visual or textual support for side views of the web interface.

2.2 Issues of Limited Working Memory Capacity and Knowledge Retrieving

When exploring unfamiliar concepts, participants frequently encounter the rabbit hole phenomenon [19, 35], engrossed in chains of nested concepts diverting them from their initial research objectives. This phenomenon is particularly prevalent in the early stages of sensemaking, as users attempt to navigate, filter, and comprehend complex topics through reading and understanding information from different resources. Additionally, this problem is compounded by context-switching between different resources or interruptions which easily happen in web searching [10]. This goal displacement disrupts the transition from information foraging to meaningful schema construction, as users become trapped in expanding information collection rather than progressing to the synthesis and organization phase of sensemaking. The challenge is further exacerbated by physical breaks between tasks, which blur previously acquired knowledge and goals, requiring additional cognitive resources to reestablish context upon returning to the task. These issues increase cognitive workload due to our limited working memory capacity. To retrieve and reuse knowledge forgotten due to this cognitive load, prior research has proposed reconstructing past contexts through *histories*, such as tracking desktop tasks [16], summarizing browsing activity [15, 23], and capturing screenshots [24, 40] to reconstruct the context of past computer work. However, these approaches primarily provide simple visual or textual histories. Additionally, personal knowledge graphs with question answering can be one way to retrieve their previous knowledge [9].

2.3 Relation between UI, Cognitive States, and Eye-tracking

Eye-tracking is a method that measures where, how long, and in what manner a person looks at a given stimulus using an eye tracker. It serves as a tangible way of representing attention. Various eye-tracking metrics [5], such as fixation, saccades, blinks, and pupillometry, provide insights into different aspects of gaze behavior. Research leveraging these eye-tracking metrics has been

conducted across multiple domains to examine selection, focus, and cognitive states. Eye-tracking has been widely utilized in cognitive-demanding fields such as software engineering [39], machine operation [50], decision making [8] and driving [56] to understand user behavior patterns and assess cognitive load. Based on these capabilities, several studies have attempted to predict users' cognitive states [6, 20, 53], while others have focused on generating scanpaths for given UI designs [26]. Data collection remains a crucial component of these modeling studies, with ongoing research exploring how to collect reliable data using simple equipment [37]. Given these characteristics, researchers have leveraged eye-tracking methods to inform user interface design and improvements [14], creating a valuable feedback loop between cognitive understanding and interaction design.

3 Attention Modeling

Gaze movement is a very natural behavior; it has been used to measure cognitive load, how users' attention goes, etc. Given this characteristic, our research team explores the way of modeling attention states through eye-tracking data, which lowers the burden that comes from trying additional behaviors such as moving to another window, expressing their thoughts as text, or being used to a transformed interface. As a goal of our system is to conserve the natural searching process as much as possible, we draft a modeling method for user attention through eye-tracking techniques, which can be fundamental to the system.

In this section, we are talking about data collection methods, creating a taxonomy, and modeling methods.

3.1 Data Collection

We have developed a protocol to collect eye movement data from participants. The protocol includes six carefully designed tasks that require multiple searching and thinking processes. Each participant is asked to answer questions related to these tasks. For each task, we provide participants with approximately 10 pre-selected websites in the form of a hyperlink list. These websites contain multiple text paragraphs with relevant information from sources such as articles and online forums.

Participants are randomly assigned two questions to answer through web searching. After task completion, we process the collected data and screen recordings into short clips of 8-10 seconds each, enhanced with scanpath visualization. This visualization helps participants recall their cognitive states during the task. We ask participants to watch these clips and assign a single attention state label to each clip based on their memory of the experience. While we provide our initial taxonomy to guide the labeling process, participants can also suggest new labels if they feel none of the existing categories adequately represent their attentional state. Further details regarding the taxonomy creation process are in Section 3.2.

For eye movement tracking, we use a monitor-mounted Tobii Fusion Pro eye tracker. Since this equipment requires a stable environment to capture gaze accurately, we conduct all data collection in a controlled lab-setting with consistent lighting. Participants are seated in non-swivel chairs and instructed to maintain a stable posture throughout the session to ensure data quality.

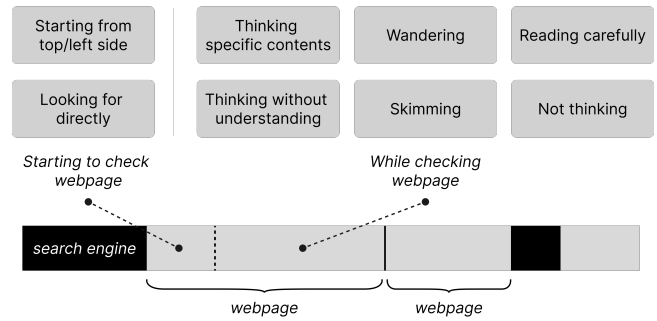


Figure 2: Initial taxonomy of attention states. It will be given to participants to facilitate the labeling process, but should be changed through data collecting process.

3.2 Creating Taxonomy

For classification tasks, a proper taxonomy is essential. For our modeling approach, we propose an initial taxonomy of eight distinct states based on how users explore web pages: two initial navigation states and six subsequent exploration states (Figure 2). While this taxonomy cannot yet fully represent users' attentional states, it will be reconstructed through a data collection study where participants directly label their own states.

To label a particular eye-tracking segment, we ask participants to consider the sequential structure of attention states shown in Figure 2. The sequential structure first distinguishes segments by whether the user is in an initial navigation phase or in a deeper exploration phase. For example, in a segment where a participant was initially scanning a page and then engaged in deep reading of specific content, the appropriate label would be "Wandering" as this represents the primary attentional state during that segment. In cases where two states are appropriate and fall under the same stage (e.g., if the participant was both reading carefully and thinking specific contents), they are asked to pick the state in which they spent the majority of the time. These issues arise because we collect a single attention state label for each eye-tracking segment.

After entering a webpage through a search engine, users typically exhibit one of two initial states depending on whether they have a clear navigational purpose: either general top-left exploration or direct navigation to specific information. Subsequently, as users explore webpage content, we categorize their behavior into six states based on comprehension level and degree of focus. These categories can be analyzed using several eye-tracking metrics outlined in Table 1.

3.3 Eye Movement Data and Modeling

The collected eye movement dataset will comprise numerous labeled segments, with each segment containing approximately 1024 data points. Instead of directly using raw x and y coordinates, we convert these data points into low-level gaze events (detailed in Table 1) that can be meaningfully linked to users' attentional states. This conversion strategy helps mitigate the challenges posed by varying UI layouts across multiple websites.

Metric	Description
Fixation [41]	Highlights sections that users selectively focus on, indicating concentrated reading behavior.
Saccade [41]	Measures gaze transition speed; rapid and repeated scanning over a broad area may indicate cognitive overload or difficulty in information processing.
Pupillometry [49]	Quantifies attention by measuring pupil dilation per unit time to identify the not-thinking state.
Blink Rate [34]	Tracks the number of blinks per unit time; a decreased blink rate is often associated with increased cognitive load and attention.

Table 1: Primary low-level gaze events and their application to attention modeling.

Table 1 describes the low-level metrics that provide insights into user engagement and facilitate the identification of key focus areas in gaze scan paths. Specifically, fixation and saccade metrics enable us to infer reading patterns that indicate deeper cognitive processing. Complementing these behavioral measures, psychophysiological metrics such as pupil dilation and blink rate offer additional physiological indicators of cognitive load and attentional shifts.

Within each webpage, we designate UI elements such as navigation bars, search fields, and interactive buttons as areas of interest (AOIs). These AOIs facilitate the measurement of gaze distribution patterns, which directly contribute to the weighting of knowledge base elements. We also incorporate the total time spent on each webpage to determine its overall significance. These computed weightings play a crucial role in constructing the personalized knowledge base described in Section 4.2.

This approach enables us to refine the importance of AOIs based on sustained focus patterns rather than isolated gaze instances. Additionally, we derive higher-level features including fixation duration distributions, saccade amplitudes, scanpath complexity metrics, and transition probabilities between AOIs to create a comprehensive feature set for our attention classification model.

4 System Design

4.1 Proactive Support based on User Attention

The automatic support powered by attention modeling in Figure 1(a) represents a significant advancement over previous practices, requiring less effort and mental workload to acquire LLMs support. For our main interface, we propose a floating window design to address key challenges in conventional LLMs interactions. Traditional interactions that involve switching windows and manually crafting prompts often disrupt the natural information-seeking flow, increasing cognitive load whereas the purpose is to reduce it. Our floating window interface minimizes context switching costs while maintaining design versatility. Industry solutions such as Arc [1] and Sider [2] have adopted similar floating interfaces to preserve

information-seeking flow. Our research extends this paradigm by implementing a proactive mechanism driven by user attention patterns. The system continuously monitors attention states during information foraging and triggers predefined support interventions at the moments. While this approach shows promise, the user perception of such interventions remains largely unexplored.

The effectiveness of proactive assistance depends critically on timing—providing support precisely when users need it most while avoiding unnecessary interruptions. To identify the optimal balance between assistance and distraction, we propose three proactive mechanisms tailored to specific attention states:

Contextual Zoom-in: This support activates when users stare at specific screen areas, interpreted as needing additional information to develop their thoughts. For example, when a user is detected fixating on a technical term like "transformer architecture" for more than 3 seconds without scrolling or clicking (indicating cognitive engagement without comprehension), the system triggers a small floating card with a simplified explanation: "Transformers are neural networks that process sequences using attention mechanisms. Need more details?" This allows users to explore topics deeper through conversation without disrupting their reading flow.

Information Pruning: This intervention responds to detected wandering attention, often indicating high mental workload. Consider a scenario where eye-tracking identifies rapid scanning across multiple sections of a dense research paper. The system highlights the three most relevant paragraphs based on previous reading patterns and search query intent, temporarily reducing the visual prominence of peripheral content. This guidance redirects focus to pertinent information, reducing cognitive load while maintaining access to the complete document.

Distraction Recovery: This support activates when detecting repetitive viewing patterns in the same areas, suggesting lost concentration or confusion. When the system observes a user repeatedly reading the same paragraph about climate modeling three times in succession, it might gently prompt: "I notice you've reviewed this section multiple times. Are you looking for something specific about climate models? Your original search was about 'effects of carbon emissions on ocean temperatures'." This contextual reminder helps users recover their original search purpose after losing focus. All support mechanisms follow a two-step approach: the system automatically suggests concise information through a floating window, and users can select whether they wish to explore further. We are planning to refine these mechanisms through iterative modeling and user feedback.

However, two significant challenges are expected. Firstly, false positive errors in attention state classification may lead to inappropriate interventions that users perceive as disruptive. We are exploring rule-based heuristics applied on top of classification results to mitigate this risk. Secondly, the displayed content may not always align with users' actual information needs. To address this challenge, we are developing a knowledge graph approach as a closer proxy for user intention and corresponding knowledge requirements.

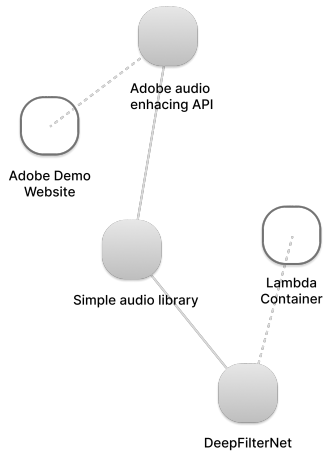


Figure 3: A knowledge graph synthesized from users' search history. It can provide an organized, linear browsing history that reflects the individual's search intent and focus at the time.

4.2 Synthesizing Knowledge Graph of Browsing History

To enable effective proactive support as described in Section 4.1, building a knowledge graph by filtering and accumulating information from browsing history can be beneficial (Figure 1(a), (d)). Moving beyond the limitations of previous less-grounded LLMs, integrating attention modeling with this graph-based knowledge base can provide more effective information retrieval. This knowledge graph can be constructed by prioritizing information based on user attention patterns. Concepts that users find confusing or significant can be emphasized through highlighting, while less relevant information can be deprioritized.

Moreover, the knowledge graph itself can be a well-organized way of representing previous browsing history while capturing user intention. Figure 3 depicted the graph. Imagine resuming a search for an audio enhancing API that you started three days or three hours ago. We typically forget the context and methods of our previous searches. Offering the relevant browsing history in visual graph form representing the relationships between explored topics can aid memory retrieval, rather than linearly listed visited website history.

However, a potential drawback of prioritizing information stems from the tendency of domain novices to oversimplify problems by selecting only a subset of features to address, causing them to miss important problem features or relationships [28, 47]. This situation could lead to knowledge graphs that are misaligned with what users actually need to acquire through the browsing process. Our research team aims to address this challenge through proper weighting between attention patterns and domain knowledge, ensuring that personalization enhances rather than limits information comprehension.

5 Conclusion

This position paper proposes an attention-driven approach to enhance sensemaking in complex online environments. By modeling user attention through eye-tracking, our system captures real-time cognitive signals that reflect users' evolving focus, intent, and comprehension. This fine-grained attention modeling serves as the foundation for context-aware support that goes beyond traditional prompt-based interactions. Building on this foundation, our system delivers proactive interventions—such as contextual explanations, content highlighting, and distraction recovery—through a lightweight floating UI that minimizes workflow disruption. These interventions are driven not by browsing history alone, but by dynamic attention patterns, allowing for more precise and cognitively aligned assistance. Furthermore, our system constructs a knowledge graph grounded in users' attention histories, enabling cross-session continuity and richer context retention. This multi-modal approach, centered on user cognition, lays the groundwork for more intuitive, responsive, and human-centered generative AI systems. Looking ahead, we plan to incorporate additional modalities such as mouse movement and biometrics, and to explore more natural interaction methods for grounding AI in users' lived workflows.

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